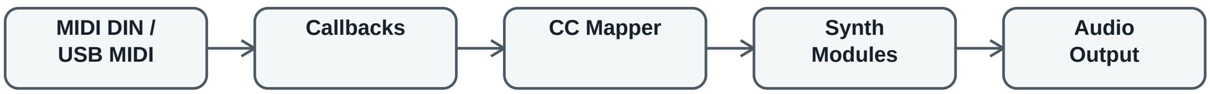
SPinSynth-T MIDI Operation Manual

MIDI and USB-MIDI control map for the Teensy monophonic virtual analog synthesizer

Scope

This manual describes how the current SPinSynth-T sketch responds to MIDI notes, pitch bend, and MIDI Control Change messages. It documents the behavior in the current code, not only the historical comments in the sketch header. The mappings are suitable for the original Arturia KeyLab-style controller setup and can also serve as a parameter dictionary for future HMI work.



MIDI Inputs

- SPinSynth-T listens to standard DIN MIDI through the MIDI library using MIDI_CHANNEL_OMNI.
- SPinSynth-T also listens to USB-MIDI through usbMIDI.
- The same note, control-change, and pitch-bend callbacks are connected to both MIDI paths.
- Compile target in the original project is Teensy 3.1/3.2 with USB Type set to Serial + MIDI.

Performance Messages

Message	Behavior
Note On	Starts oscillator note tracking, triggers the filter envelope, opens filter gate state, triggers the amplifier envelope, and opens amplifier gate state.
Note Off	Releases the matching key event. The oscillator keeps monophonic highest-note priority behavior, while filter and amplifier gates receive note-off.
Pitch Bend	Sent directly to the oscillator. Positive bend raises pitch and negative bend lowers pitch according to oscillator pitch-bend logic.

Global Controls

CC	Name	Value Range	Effect
1	Mod Wheel	0-127	Controls the currently selected mod-wheel destination: filter cutoff, filter resonance, filter LFO amount, or oscillator LFO amount.
7	Master Volume	0-127	Sets amplifier master volume.
18	Mod Wheel Function	See table below	Selects what CC#1 will control.
35	Portamento	0-127	Sets oscillator portamento/glide amount.
41	Master Tune	0-127	Sets oscillator master tuning. Value 63 is center/near A=440 reference.

Mod Wheel Function Selector CC#18

CC#18 Value	Selected CC#1 Destination
0-20	Filter cutoff modulation
22-42	Filter resonance
43-62	Filter LFO amount
63-118	Oscillator LFO amount
121-127	Oscillator LFO amount

Note: the current code preserves small value gaps from the original mapping: 21, 119, and 120 do not change the selected function.

Oscillator Controls

CC	Name	Value Range	Effect
16	Oscillator LFO Rate	0-127	Sets oscillator LFO rate. Current code maps value / 10.0, approximately 0.0 to 12.7 Hz.
17	Oscillator LFO Amount	0-127	Sets oscillator LFO amplitude/depth.
37	Saw X-Factor	0-127	Sets detuned saw factor in cents around center. Used by XTREME detuned saw behavior.
38	Pulse Width	0-127	Sets pulse width directly. Higher values reduce pulse width from center toward a narrower pulse.
39	Sub Factor	0-127	Sets sub oscillator factor from 0.5 upward. Used by the XTREME sub oscillator.
40	Oscillator Waveform	See waveform table	Selects SINE, SAW, SQUARE/PULSE, variable pulse width, or XTREME.
42	Saw Amount	0-127	Sets XTREME saw component amount.
43	Pulse Amount	0-127	Sets XTREME pulse component amount.
44	Sub Amount	0-127	Sets XTREME sub oscillator amount.
91	Oscillator LFO Waveform	See LFO waveform table	Selects oscillator LFO waveform and pulse width region.

Oscillator Waveform Selector CC#40

CC#40 Value	Waveform
0-20	SINE
22-42	SAW
43-62	SQUARE_PULSE with centered pulse width
63-118	SQUARE_PULSE with variable pulse width
121-127	XTREME with centered pulse width

Note: values 21, 119, and 120 are gaps in the current range checks and do not select a new waveform.

Pulse And XTREME Operating Notes

- SQUARE_PULSE is produced by subtracting two saw-derived phases, so pulse width changes the phase distance between the two edges.
- CC#38 directly adjusts pulse width for SQUARE_PULSE and XTREME modes.
- XTREME combines main saw, detuned saw, pulse, and sub components. CC#42, CC#43, and CC#44 control the saw, pulse, and sub mix amounts.
- CC#37 changes the detuned saw factor around the center value, while CC#39 changes the sub oscillator factor.

LFO Waveform Selectors CC#91 And CC#93

Value	LFO Waveform
0-16	SINE
17-32	SAW
33-48	INVERTED_SAW
49-64	SQUARE_PULSE with centered pulse width
65-127	SQUARE_PULSE with variable pulse width

Filter Controls

CC	Name	Value Range	Effect
19	Filter Envelope Level	0-127	Sets filter envelope amount and polarity around the middle range. Values below center invert the envelope; values above center apply normal envelope modulation.
71	Filter Resonance	0-127	Sets filter resonance, with internal limiting for stability.
72	Filter Release	0-127	Sets filter envelope release time.
73	Filter Attack	0-127	Sets filter envelope attack time.
74	Filter Cutoff	0-127	Sets main filter cutoff with an exponential response for finer low-frequency control.
75	Filter Decay	0-127	Sets filter envelope decay time.
76	Filter LFO Rate	0-127	Sets filter LFO rate. Current code maps value / 10.0, approximately 0.0 to 12.7 Hz.
77	Filter LFO Amount	0-127	Sets filter LFO amount. Current code maps value / 511.0 for a restrained modulation range.
79	Filter Sustain	0-127	Sets filter envelope sustain level.
93	Filter LFO Waveform	See LFO waveform table	Selects filter LFO waveform and pulse width region.

Filter Control Response

- CC#74 is exponential rather than linear. This gives more usable resolution in the lower cutoff range, where small changes are musically important.
- CC#19 is bipolar. Around the middle of the control the filter envelope amount is close to zero. Lower values create inverted envelope modulation, and higher values create normal positive envelope modulation.
- The CC#19 response is curved on both sides of center, giving finer adjustment near zero while preserving the same maximum negative and positive depth at the extremes.
- Resonance and cutoff are internally limited to keep the filter stable.

Amplifier Controls

CC	Name	Value Range	Effect
80	Amplifier Attack	0-127	Sets amplifier envelope attack time.
81	Amplifier Decay	0-127	Sets amplifier envelope decay time.
82	Amplifier Sustain	0-127	Sets amplifier envelope sustain level.
83	Amplifier Release	0-127	Sets amplifier envelope release time.

Important: the historical sketch header mentions amplifier ADSR on CC#87-90. The current code uses CC#80-83. This manual follows the current code.

Envelope Value Behavior

- Attack values are converted to attack time by the EnvelopeGenerator using (value + 1) * 10.0.
- Decay values are converted using value * 100.0 + 1.0.
- Sustain values map directly to value / 127.0.
- Release values are converted using (value + 1) * 100.0.
- Envelope curves are exponential rather than simple linear ramps.

Practical Starting Points

Goal	Suggested MIDI Setup
Basic saw lead	CC#40 in SAW range, CC#74 around middle/high, CC#71 low to medium, amp ADSR short attack and medium sustain.
Pulse tone	CC#40 in SQUARE_PULSE range, adjust CC#38 for pulse width, use CC#74/71 for brightness and edge.
XTREME texture	CC#40 in XTREME range, raise CC#42/43/44 gradually, tune CC#37 and CC#39 by ear.
Filter sweep	Use CC#74 for base cutoff, CC#19 above center for positive envelope amount, and CC#73/75/79/72 for envelope shape.
Inverted filter envelope	Set CC#19 below center, then adjust CC#74 so the inverted movement has enough room to be heard.

Suggested Controller Layout

Control Area	Recommended Assignment
Performance	Keyboard note on/off, pitch bend, mod wheel CC#1
Global	Master volume CC#7, master tune CC#41, mod-wheel function CC#18, portamento CC#35
Oscillator	CC#16, 17, 37, 38, 39, 40, 42, 43, 44, 91
Filter	CC#19, 71, 72, 73, 74, 75, 76, 77, 79, 93
Amplifier	CC#80, 81, 82, 83

Operational Notes

- Use MIDI monitor tools or the controller editor to verify that the controller sends the CC numbers expected by this manual.
- For waveform and mod-wheel selector controls, avoid the documented gap values if deterministic selection is needed.
- The modulation wheel destination remains whatever was last selected by CC#18.
- The synth is currently monophonic, with note priority handled inside the oscillator key-event logic.
- The code listens to all MIDI channels on DIN MIDI; USB-MIDI callbacks do not filter channels in the sketch.